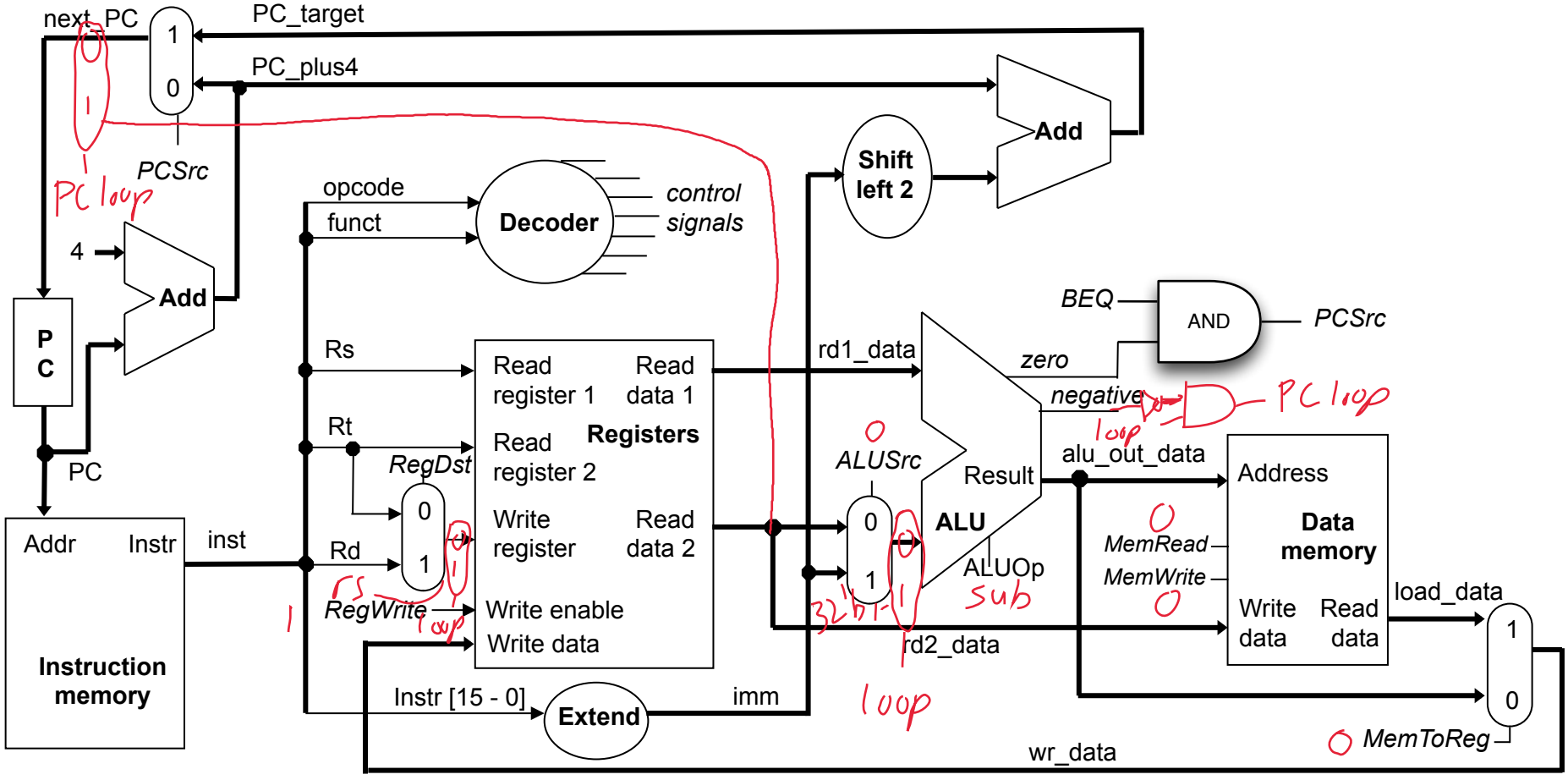

next PC
1
0
PC_target
PC_plus4
PCSrc
PC loop
4
Add
P
C
PC
Addr
Instr
inst
Instruction memory
opcode
funct
Decoder
control signals
Rs
Rt
Rd
RegDst
RegWrite
Read register 1
Read register 2
Registers
Write register
Write data
Read data 1
Read data 2
Write enable
Write data
Instr [15 - 0]
Extend
imm
Shift left 2
Add
BEQ
AND
PCSrc
rd1_data
zero
negative
loop
PC loop
ALUSrc
Result
ALU
ALUOp
sub
rd2_data
32'h1-1
loop
alu_out_data
Address
Data memory
MemRead
MemWrite
Write data
Read data
load_data
MemToReg
wr_data